



AnalystLab

Africa - Data

Science Internship Programme

Week 3 Assignment

Statistical Analysis Report

**Advanced Data Exploration & Feature Engineering
Loan Prediction Dataset**

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Analytical scope

This report evaluates the statistical evidence behind the relationships observed during exploratory analysis of 614 loan applications. Six hypothesis tests cover categorical associations, differences between approved and rejected groups, geographic income variation and the relationship between household income and requested loan amount.

Decision framework

Analytical element	Application
Significance threshold	alpha = 0.05
Categorical association	Chi-square test of independence
Two skewed numerical groups	Mann-Whitney U test
Three skewed numerical groups	Kruskal-Wallis test
Monotonic numerical relationship	Spearman rank correlation
Categorical effect size	Cramer's V

Assumption assessment

Income and loan variables were strongly right-skewed and contained valid extreme observations. Normality checks and visual inspection therefore supported nonparametric rank-based comparisons. For chi-square testing, contingency-table expected counts exceeded common minimum thresholds, supporting the use of the approximation.

Interpretation rule A small p-value supports an association or group difference; it does not establish causation, business importance or fairness.

Test 1: Credit history and loan approval

Objective: Determine whether credit-history status and loan approval are associated.

Null hypothesis (H0): Credit history and loan approval are independent.

Alternative hypothesis (H1): Credit history and loan approval are associated.

Credit history	Rejected	Approved	Approval rate
0 - No positive history	82	7	7.87%
1 - Positive history	110	415	79.05%

Result: chi-square = 176.115; degrees of freedom = 1; $p = 3.418e-40$; Cramer's $V = 0.536$; minimum expected count = 27.83.

Decision and implication: Reject H0. Credit history has a strong, statistically supported relationship with historical approval and should remain a core modelling feature.

Test 2: Property area and loan approval

H0: Property area and approval are independent. H1: Property area and approval are associated.

Property area	Rejected	Approved	Approval rate
Semiurban	54	179	76.82%
Urban	69	133	65.84%
Rural	69	110	61.45%

Result: chi-square = 12.298; degrees of freedom = 2; $p = 0.0021$; Cramer's $V = 0.142$. Reject H0: property area is significant, but its effect is small relative to credit history.

Test 3: Total income by approval outcome

Objective: Compare the distributions of total household income for approved and rejected applications.

H0: Total-income distributions are equivalent across approval outcomes. H1: The distributions differ.

Outcome group	Observations	Median total income
Approved	422	5,439.00
Rejected	192	5,289.50

Result: Mann-Whitney U = 41,262.5; p = 0.7128; rank-biserial effect = -0.019. Fail to reject H0.

Business implication: Total household income does not independently distinguish approval outcomes in this sample. Income may still matter when combined with requested loan amount, repayment period or credit history.

Test 4: Loan amount by approval outcome

H0: Requested loan-amount distributions are equivalent across approval outcomes. H1: The distributions differ.

Outcome group	Observations	Median loan amount
Approved	422	128.00
Rejected	192	128.00

Result: Mann-Whitney U = 38,788.0; p = 0.3976; rank-biserial effect = 0.043. Fail to reject H0.

Business implication: Loan size should not be interpreted in isolation; affordability-oriented combinations are more defensible candidates for modelling.

Test 5: Total income across property areas

Objective: Assess whether the observed property-area approval pattern could be explained by materially different household-income distributions.

H0: Rural, urban and semiurban applicants have equivalent total-income distributions. H1: At least one group has a different distribution.

Property area	Observations	Median total income
Rural	179	5,704.00
Semiurban	233	5,191.00
Urban	202	5,258.50

Result: Kruskal-Wallis $H = 3.933$; $p = 0.1400$. Fail to reject H0.

Business implication: The geographic approval differences cannot be attributed solely to statistically distinguishable total-income distributions in this dataset.

Test 6: Income and requested loan amount

H0: There is no monotonic relationship between total household income and requested loan amount.

H1: A monotonic relationship exists.

Result: Spearman $\rho = 0.688$; $p = 3.423e-87$. Reject H0.

Business implication: Higher-income applicants generally request larger loans. This relationship supports the construction of repayment-burden and income-relative features, but does not show that income causes approval.

Consolidated statistical results

Analysis	Statistic	p-value	Decision
Credit history vs approval	176.11	<0.001	Reject H0
Property area vs approval	12.30	0.002	Reject H0
Total income by approval	41,262	0.713	Do not reject
Loan amount by approval	38,788	0.398	Do not reject
Income across property areas	3.93	0.140	Do not reject
Income vs loan amount	0.688	<0.001	Reject H0

Effect size and practical significance

Credit history combines a very small p-value with a comparatively strong effect size. Property area is statistically significant but has a weaker effect, so it should not receive equal operational weight. The Mann-Whitney comparisons have small rank-biserial effects, reinforcing the conclusion that raw income and loan size alone offer limited separation.

Limitations and interpretation controls

- o Multiple tests were conducted; borderline results should be interpreted cautiously and confirmed in model validation.
- o The target reflects historical approvals rather than repayment or default.
- o Non-significance does not prove that a feature has no predictive value in interaction with others.
- o Statistical associations do not establish causal or legally permissible lending criteria.

Conclusion

The statistical evidence supports prioritizing credit history, retaining property area as a secondary feature, engineering income-relative affordability indicators and validating all conclusions using out-of-sample modelling and fairness checks.